



Mechanical, Industria... > MIAE 215 Midterm Cheatsheet: Control Statements & Data Types

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```
for(i=0;k<5;i++) C[i] = A[i];
```

```
// append the elements of B onto the end of C  
for(i=0;k<5;i++) C[nA + i] = B[i];
```

```
// don't forget the termination character  
C[nA+nB] = '\0';
```

Consider a $n \times n$ matrix A where n is an integer between 1 and 10. Therefore, you should declare a 2D array A with a size of 10×10 . Develop a program that:

- Inputs the matrix dimension n from the keyboard.
- Checks if the value of n is between 1 and a maximum of 10. If the input value is outside the valid range then set $n = 10$, and print an error message to the screen that: "Invalid input, n must be between 1 and 10".
- Initializes the elements of the matrix to $A[i][j] = -2.0^i$.
- Calculates (and prints to the screen) the minimum and maximum of $A[i][j]$, and the number of times $A[i][j]$ is equal to zero.

```
int main()
{
    int i, j, n, count;
    double A[10][10] = {0.0}; // Initialize A to zero
    // note: a matrix starts at A[1][1] and goes to a maxi
    // note: however, we can only use A[1][1] to A[n][n] if n
    double min, max;
    a)
    cin >> n;
    b)
    if(n < 1) {
        cout << "Invalid input out of bounds";
        n = 1;
    }
    if(n > 10) {
        cout << "Invalid input out of bounds";
        n = 10;
    }
    c)
    for(i=1; i<=n; i++) {
        for(j=1; j<=n; j++) {
            A[i][j] = -2.0^i + j^j;
        }
    }
    d)
    min = max = A[1][1];
    count = 0;
    for(i=1; i<=n; i++) {
        for(j=1; j<=n; j++) {
            if(A[i][j] > max) max = A[i][j];
            if(A[i][j] < min) min = A[i][j];
            if (abs(A[i][j]) < 1.0e-10) count++;
        }
    }
    cout << "min = " << min;
    cout << "max = " << max;
    cout << "number of 0.0 = " << count;
}
```

Mock exam

Summary

Quiz

Ask a question about this document

